

Microprocessors And Microcontrollers

Assignment 3- Week 3

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15

QUESTION 1:

Correct Answer: b

Detailed Solution:

1. `XRA A`: This instruction XORs register A with itself, effectively clearing register A (set to 0).
2. `MVI C, 05H`: This instruction loads register C with the value 05H (5 in decimal).
3. `LOOP: DCR C`: This instruction decrements the value in register C by 1.
4. `JNZ LOOP`: This instruction jumps back to `LOOP` if the Zero flag is not set (i.e., if C is not zero).

The loop will start with C = 5. Each time through the loop, C is decremented, and the loop continues as long as C is not zero. Therefore:

- First iteration: C = 5 → decremented to 4.
- Second iteration: C = 4 → decremented to 3.
- Third iteration: C = 3 → decremented to 2.
- Fourth iteration: C = 2 → decremented to 1.
- Fifth iteration: C = 1 → decremented to 0.

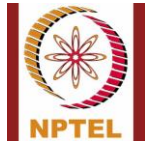
When C becomes 0, the `JNZ` instruction will fail (because the Zero flag is set), and the loop will exit.

Thus, the loop will be executed **five times**.

QUESTION 2:

Correct Answer: d

Detailed Solution:



In the 8085 microprocessor, the flags (Zero, Carry, Sign, and others) can be affected by various instructions, including conditional branch instructions. However, the specific behavior of conditional branch instructions depends on the flags that they check. Here's a breakdown:

- **Zero flag (Z):** The Zero flag is checked and affected by instructions like `JZ` (Jump if Zero) or `JNZ` (Jump if Not Zero). These instructions make decisions based on the status of the Zero flag, so it is not unaffected.
- **Carry flag (C):** The Carry flag is checked and affected by conditional branch instructions like `JC` (Jump if Carry) or `JNC` (Jump if No Carry), meaning it is also not unaffected.
- **Sign flag (S):** The Sign flag, which indicates the sign of the result in the accumulator, is also checked by conditional branch instructions like `JS` (Jump if Sign) or `JNS` (Jump if No Sign).
- **None of the given options:** This means that all of the listed flags (Zero, Carry, Sign) can be affected by conditional branch instructions. Therefore, none of the flags in the given options is unaffected.

QUESTION 3:

Correct Answer: a

Detailed Solution:

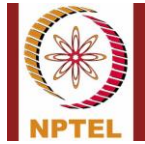
TRAP is non maskable interrupt, but other interrupts can be maskable using interrupt enable(IE).

QUESTION 4:

Correct Answer: d

Detailed Solution:

Whenever a interrupt signal encountered by MP, then MP stops execution of the current instruction and start service the interrupt.



QUESTION 5:

Correct Answer: a

Detailed Solution:

1. MVI A, 65H

This instruction loads the accumulator (register A) with the value **65H** (101 in decimal). So, at this point:

- **A = 65H**

2. MVI B, 32H

This instruction loads register B with the value **32H** (50 in decimal). So, at this point:

- **B = 32H**

3. CMP B

The **CMP** instruction compares the contents of the accumulator (A) with the contents of register B. It performs the operation:

- **A - B**
- This operation is performed without modifying the accumulator. Instead, it updates the flags based on the result of the subtraction.
- **65H (A) - 32H (B) = 33H**

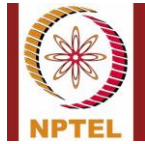
Since the result of the subtraction is **33H**, which is a positive value (non-zero), the **Zero flag** will be **reset** (set to 0) because the result is non-zero. The contents of the accumulator (A) remain **65H**, since **CMP** does not modify the accumulator.

Thus, after the execution:

- **Zero flag = 0** (because $A \neq B$)
 - **Accumulator contents = 65H** (unchanged after **CMP**)
-

QUESTION 6:

Correct Answer: c



Detailed Solution:

1. **MVI A, 35H**

- The **MVI** instruction (Move Immediate) takes 2 bytes: 1 byte for the opcode and 1 byte for the immediate data.
- This instruction occupies **2 bytes**.

2. **MVI B, 23H**

- Similar to the previous instruction, **MVI B, 23H** also occupies **2 bytes** (1 byte for the opcode and 1 byte for the immediate data).

3. **ADD B**

- The **ADD** instruction (Add Register) occupies **1 byte** because it only requires 1 byte for the opcode (since it adds the contents of register B to the accumulator).

Now, let's calculate the total size:

- **MVI A, 35H** = 2 bytes
- **MVI B, 23H** = 2 bytes
- **ADD B** = 1 byte

Total = 2 + 2 + 1 = **5 bytes**

QUESTION 7:

Correct Answer: c

Detailed Solution:

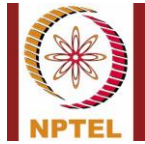
• **a) Stack is a last-in-first-out structure.**

- This is **true**. The stack in the 8085 microprocessor follows the **LIFO** (Last-In, First-Out) principle, meaning the last item pushed onto the stack is the first one to be popped off.

• **b) Information is saved on the stack by pushing on it.**

- This is **true**. The **PUSH** instruction is used to save information onto the stack, which involves writing data to the memory location pointed to by the stack pointer (SP).

• **c) Size of the register associated with the stack is 8 bits.**



- This is **false**. The **stack pointer (SP)** is a **16-bit register**, not an 8-bit register. It holds the address of the current top of the stack and allows for 16-bit memory addressing.
- **d) Information is retrieved on the stack by popping it off.**
 - This is **true**. The **POP** instruction is used to retrieve data from the stack by removing the topmost element and adjusting the stack pointer.

QUESTION 8:

Correct Answer: d

Detailed Solution:

- **a) LXI SP, 0FFFH**
 - This instruction loads the stack pointer (SP) with the address **0FFFH**. It is a valid operation, setting the SP to point to a specific memory address.
- **b) PUSH PSW**
 - The **PUSH PSW** instruction is valid. It pushes the contents of the **PSW (Program Status Word)** onto the stack. The PSW contains the accumulator and the flag register, which are stored in memory when this instruction is executed.
- **c) SPHL**
 - The **SPHL** instruction is valid. It copies the contents of the **HL** register pair to the stack pointer (SP), setting the SP to the address in HL. This operation is often used to manipulate the stack pointer.
- **d) POP C**
 - This instruction is **invalid**. The **POP** instruction is used to retrieve data from the stack, but it requires the use of a **16-bit register pair** to store the popped data. Register C is an **8-bit** register, so it cannot be used with the **POP** instruction. For example, valid **POP** instructions would be **POP BC**, **POP DE**, etc., which use 16-bit register pairs.

QUESTION 9:



Correct Answer: d

Detailed Solution:

This is **false**. The first `RET` instruction must always correspond to a previous `CALL` instruction. If a `RET` appears before a `CALL`, there is no return address on the stack, leading to undefined behavior or an error in program execution.

QUESTION 10:

Correct Answer: d

Detailed Solution:

- `MVI C, 03H`: Load register C with 03H.
- `LXI H, 2000H`: Load the HL register pair with 2000H. This means H = 20H and L = 00H.
- `MOV A, M`: Move the contents of the memory location pointed to by HL (2000H) into register A. The value at memory location 2000H is 18H.
Now, A = 18H.
- `DCR C`: Decrement register C.
C = 03H → 02H.

Loop - L1 (First Iteration)

- `L1: INX H`: Increment the HL register pair. Now HL = 2001H.
- `MOV B, M`: Move the contents of memory location 2001H into register B. The value at 2001H is 10H.
Now, B = 10H.
- `CMP B`: Compare the contents of the accumulator (A = 18H) with B (B = 10H).
A - B = 18H - 10H = 08H. Since the result is not zero, the Zero flag is not set.
- `JNC L2`: Since the Zero flag is not set, the jump does not occur, and we proceed to the next instruction.
- `MOV A, B`: Now, move the contents of register B (which is 10H) into register A.
A = 10H.
- `L2: DCR C`: Decrement register C.
C = 02H → 01H.
- `JNZ L1`: Since C is not zero, we jump to L1.

Loop - L1 (Second Iteration)

- `L1: INX H`: Increment the HL register pair. Now HL = 2002H.



- MOV B, M: Move the contents of memory location 2002H into register B. The value at 2002H is 2BH.
Now, B = 2BH.
- CMP B: Compare the contents of the accumulator (A = 10H) with B (B = 2BH).
 $A - B = 10H - 2BH = -1BH$. Since the result is not zero, the Zero flag is not set.
- JNC L2: Since the Zero flag is not set, the jump does not occur, and we proceed to the next instruction.
- MOV A, B: Now, move the contents of register B (which is 2BH) into register A.
A = 2BH.
- L2: DCR C: Decrement register C.
 $C = 01H \rightarrow 00H$.
- JNZ L1: Since C is now zero, the loop terminates and we exit to the next instruction.

Post-Loop Operations

- STA 2100H: Store the contents of register A (which is 2BH) into memory location 2100H.
- HLT: Halt the program.

Final Register Contents

- A = 2BH
- B = 2BH
- C = 00H

Thus, the final contents of registers A, B, and C are 2BH, 2BH, and 00H, respectively.

QUESTION 11:

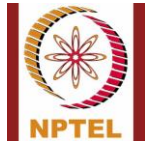
If an 8085 microprocessor works at a frequency of 1 MHz, determine the total delay of the following sequence of instructions (in seconds).

MVI A, 0FH

MVI B, 0FH

L1: DCR B

JNZ L1



- a) 218 μ s
- b) 219 μ s
- c) 220 μ s
- d) 221 μ s

Correct Answer: a

Detailed Solution:

Program Breakdown:

1. **MVI A, 0FH**
 - This instruction moves the immediate value **0FH** (15 in decimal) into register **A**.
 - The instruction takes **1 machine cycle**, which corresponds to **4 T-states**.
2. **MVI B, 0FH**
 - This instruction moves the immediate value **0FH** (15 in decimal) into register **B**.
 - The instruction takes **1 machine cycle**, which corresponds to **4 T-states**.
3. **L1: DCR B**
 - This instruction decrements the contents of register **B**.
 - **DCR** takes **1 machine cycle**, corresponding to **4 T-states**.
4. **JNZ L1**
 - This instruction checks if the **Zero flag** is **not** set and jumps back to **L1** if the condition is true.
 - **JNZ** takes **2 machine cycles** (since it may jump), corresponding to **10 T-states**.

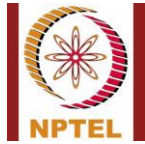
Calculation of Total Delay:

Total Delay for Each Instruction:

1. **MVI A, 0FH** takes **4 T-states**.
2. **MVI B, 0FH** takes **4 T-states**.
3. **DCR B** takes **4 T-states**.
4. **JNZ L1** takes **10 T-states**.

The loop runs until **B** reaches 0. Initially, **B = 0FH** (15 in decimal), and **DCR B** will decrement **B** by 1 each time. The loop will execute 15 times because **B** starts from 15 and is decremented until it reaches 0.

- **Total time for one loop iteration = 4 (DCR B) + 10 (JNZ L1) = 14 T-states.**



Since the loop runs 15 times, the total time for the loop is:

- Total loop time = 15 iterations $\times$ 14 T-states = 210 T-states.

Total Time for the Program:

- Total time = Time for `MVI A, 0FH` + Time for `MVI B, 0FH` + Total loop time.
- Total time = 4 (MVI A) + 4 (MVI B) + 210 (loop) = 218 T-states.

Now, let's calculate the delay in seconds.

Calculation of Time in Seconds:

- The 8085 microprocessor works at 1 MHz, which means the time for each T-state is:

$$\text{Time per T-state} = \frac{1}{\text{Frequency}} = \frac{1}{1,000,000} \text{ seconds} = 1 \mu\text{s}.$$

- Therefore, the total delay in microseconds is:

$$\text{Total delay} = 218 \times 1 \mu\text{s} = 218 \mu\text{s}.$$

QUESTION 12:

Correct Answer: b

Detailed Solution:

- `LXI 3FFFH`:
 - This instruction loads the stack pointer (SP) with the value 3FFFH.
 - So, after this instruction, SP = 3FFFH.
- `PUSH B`:
 - The `PUSH` instruction is used to push the contents of the register pair (in this case, the B register pair) onto the stack.
 - The B register pair contains the contents of registers B and C, and `PUSH B` stores these two 8-bit values onto the stack.



- Each PUSH instruction causes the stack pointer (SP) to be decremented by 2 because it is pushing two 8-bit values (16 bits total).
- So, after executing `PUSH B`, SP is decremented by 2:
 $SP = 3FFFH - 2 = 3FFDH.$
- HLT:
 - This instruction halts the program and does not affect the SP.

QUESTION 13:

Correct Answer: c

Detailed Solution:

The Program Status Word (PSW) in the 8085 microprocessor is a 16-bit register that contains two parts:

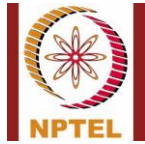
1. Accumulator (8 bits)
 2. Flag Register (8 bits)
- The Accumulator holds the result of arithmetic and logical operations.
 - The Flag Register contains the flags (Zero, Sign, Parity, Carry, and Auxiliary Carry), which indicate the status of the last operation.

Thus, the PSW is represented as the Accumulator and Flag Register pair.

QUESTION 14:

The total number of memory accesses involved (including opcode fetch), when an 8085 microprocessor executes the instruction **STA 2050H**, is:

- a) 1
- b) 2
- c) 3
- d) 4



Correct Answer: d

Detailed Solution:

The **STA 2050H** instruction stores the contents of the accumulator into the memory location 2050H. The execution involves the following steps:

1. **Opcode Fetch:** The microprocessor fetches the opcode for the STA instruction from memory. (1 memory access)
2. **Operand Fetch (Low Byte):** The lower-order byte of the address (50H) is fetched from memory. (1 memory access)
3. **Operand Fetch (High Byte):** The higher-order byte of the address (20H) is fetched from memory. (1 memory access)
4. **Memory Write:** The contents of the accumulator are written to the memory location 2050H. (1 memory access)

Total Memory Accesses: $1+1+1+1=4$

QUESTION 15:

Correct Answer: d

Detailed Solution:

- **ADD C** directly refers to the **C** register as an operand, meaning the operand is in a register.
- The operation does not involve any memory address or an immediate value but instead directly uses the contents of the **C** register.

This is an example of **register addressing mode**, where the operand is specified by a register.

*****END*****